
GIT PUSH THE BEATS

Melody Dive Use-Case Specification: Download Songs

Version 1.0

Melody Dive	Version: 1.0
Use-Case Specification: Download songs	Date: 15/11/2025
UC Download songs	

Revision History

Date	Version	Description	Author
15/11/2025	1.0	Add details information about this use case	Ngo Tran Quang Dat

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1. Use-Case Name: Download Songs

1.1 Brief Description

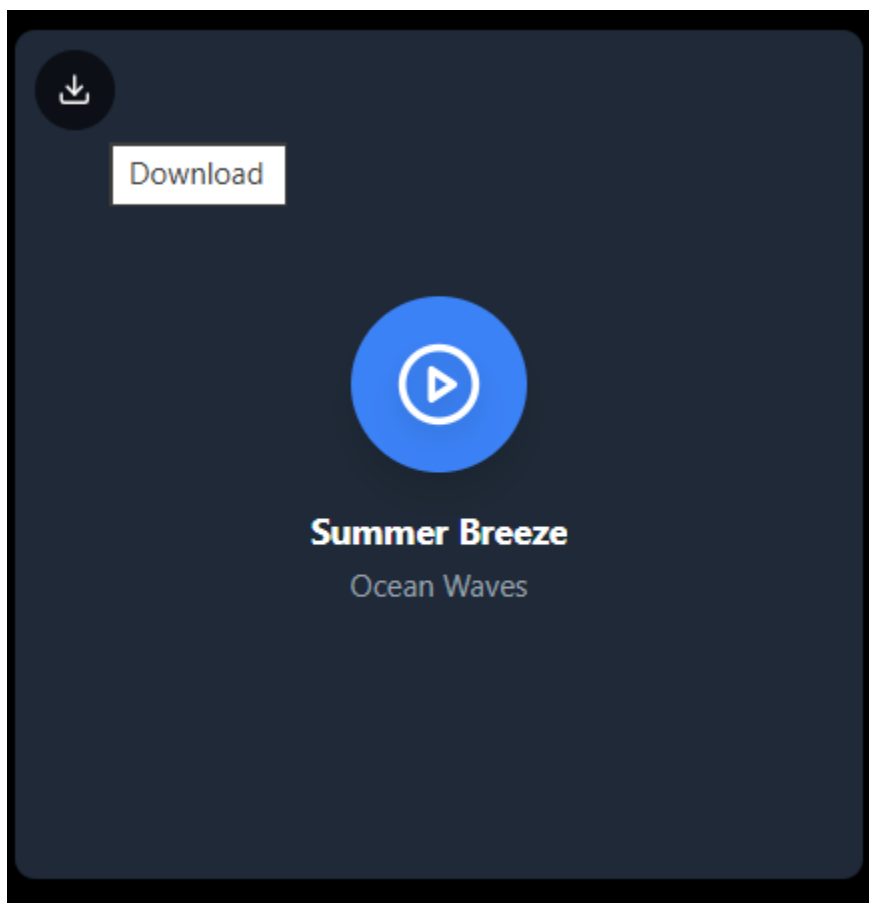
This use case describes the process allowing a Registered User to download music tracks to their local device for offline playback. Its primary purpose is to enable users to consume audio content without an active internet connection

2. Flow of Events

2.1 Basic Flow

This use case starts when the user decides to save a track for offline listening.

1. The user navigates to a song or album view, selects a specific track, and clicks the "Download" button/icon.



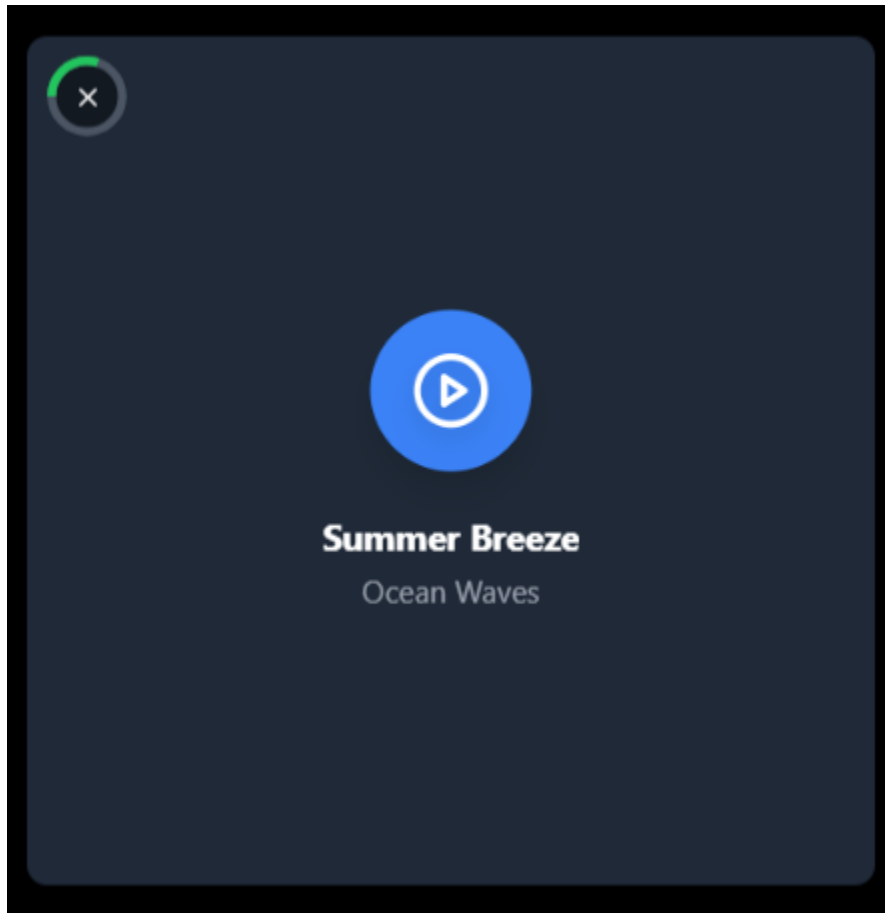
2. The system identifies the Content Classification (Free or Premium) of the selected track and retrieves the user's current Subscription Status.

3. The system verifies that the user is authorized to download the content (This authorization is valid if the User is "Active Premium," OR if the Content Classification is "Free")

4. The system checks the availability of Device Storage to ensure there is sufficient space for the file.

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5. The system initiates the secure transfer of the Encrypted Audio File to the user's local device storage.
6. The system displays a Download Progress Indicator to the user.



7. Upon successful transfer, the system updates the song's status icon to "Downloaded" and adds the track metadata to the user's "Offline Library" list.

2.2 Alternative Flows

2.2.1 Network Connection Interrupted

1. The system detects that the network connection has been lost before the file transfer is complete.
2. The system pauses the download process and temporarily saves the partial data (if supported).
3. The system updates the status of the track icon to "Paused/Error" and displays a notification: "Download failed. Please check your connection."
4. The Registered User restores the internet connection and clicks the "Retry" button (or the specific track icon).
5. The system verifies the network connection is stable.
6. The system resumes the transfer of the Encrypted Audio File from the last saved point (or restarts it if partial saving is not supported).
7. The flow of events resumes at Step 6 of the Basic Flow (Displaying Progress Indicator).

2.2.2 User Cancels the Download

This alternative flow starts during Step 6 of the Basic Flow (while the Download Progress Indicator is displayed).

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1. The user clicks the "Cancel" (or "X") button on the progress indicator.
2. The system immediately terminates the secure file transfer process.
3. The system deletes any partial temporary files associated with this track from the device storage to free up space.
4. The system reverts the song's status icon from the progress bar back to the initial "Download" state.
5. The use case ends.

3. Special Requirements

3.1 Performance Requirement (Concurrency)

The System must support asynchronous background downloading. The user must be able to continue browsing or streaming music without interruption while the download takes place. To prevent network congestion, the System limits concurrent downloads to a maximum of 3 tracks simultaneously; additional requests must be queued.

3.2 Storage Constraint

The System must perform a storage check before and during the download. If the device's free storage drops below a critical threshold (e.g., 50MB), the System must automatically pause the download and alert the user to prevent system instability.

4. Preconditions

4.1 User account logged in

The user must have logged in their account.

4.2 Content Displayed

The System is currently displaying the details of a song, album, or playlist where the "Download" option is visible.

5. Postconditions

5.1 Download Completed (Success)

The encrypted audio file is securely stored on the user's local device. The specific track is added to the "Offline Library," and the track's interface updates to display a "Downloaded" indicator (allowing offline playback).

5.2 Download Interrupted (Alternative)

The system holds a partial temporary file (if supported), and the track's interface displays a "Paused" or "Error" status, waiting for the user to resume the process.

5.3 State Unchanged (Failure/Cancellation)

No new audio files are stored on the device. The track's interface remains in (or reverts to) the initial "Download" state. This occurs if the user cancels the operation, lacks the required permissions (e.g., Normal user trying to download Premium content), or if the device storage is full.

6. Extension Points

6.1 Subscription Verification

Occurs at Step 3 of the Basic Flow, when the system determines the user does not have the required permissions (e.g., a normal user attempting to download a Premium song). The "Upgrade to Premium" use case may extend the behavior at this point to prompt a subscription purchase.

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6.2 Storage Management

Occurs at Step 4 of the Basic Flow, if the system detects insufficient device storage to save the file. The "Delete Songs" or "Manage Storage" use case may extend the behavior here to allow the user to free up space immediately.